# Stability analyis of a class of 2 x 2 triangular hyperbolic systems


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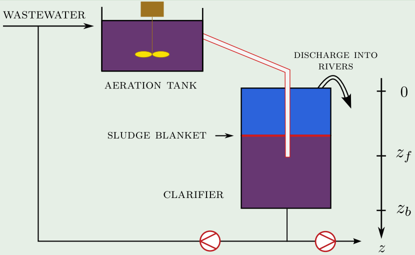
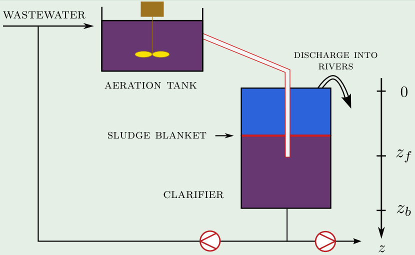
**In Collaboration with:**
Jean Auriol (L2S)

## Framework


WASTEWATER
AERATION TANK
DISCHARGE INTO RIVERS
SLUDGE BLANKET
CLARIFIER
0
$z_f$
$z_b$
$z$

## Framework




## 1D-Dynamic Model of the Clarifier (Two-Phase flow)

State variables: $\varepsilon_s$ (particles density), $f_s := \varepsilon_s v_s$ (momentum) with $v_s$ the particles velocity

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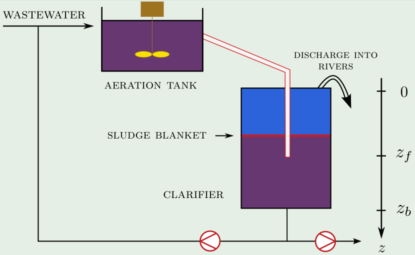


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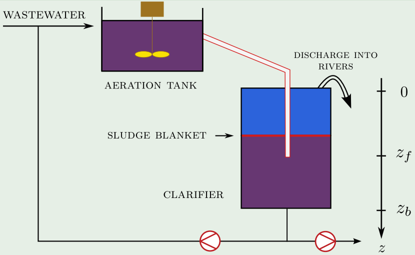


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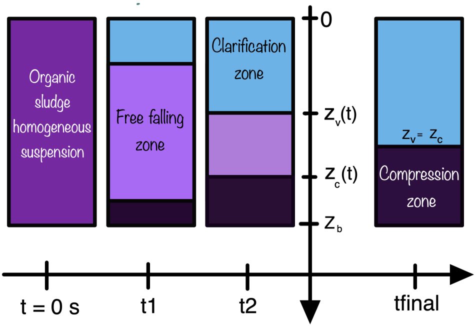
**Goal 1: Design a robust boundary controller that takes into account the Clarifier/Aeration Tank configuration (with disturbance inputs) in order to stabilize the sludge blanket depth**

**Goal 2: Prove the stability of the controlled system**

## Outline of Talk

# Batch Column Framework

# Settling in a batch column





- N. Chassin, C. Valentin, J.-M. Choubert, F. Couenne, C. Jallut, 2022, *How to improve modeling of urban sludge settling in a batch column*, Note interne
- C. Valentin, F. Lagoutière, J.-M. Choubert, F. Couenne, C. Jallut, 2023, *Knowledge-based model and simulations to support decision making in Wastewater Treatment*, Processes PROCEEDINGS 33rd European Symposium on Computer Aided Process Engineering (ESCAPE33), (6p), Athens. Greece

## System analysis

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$\sigma_0, \ n_s, \ n_r, \ \varepsilon_c \ , \ \tau, \ \rho_s$ (parameters)

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Eigenvalues of **A(Y)** ?

$$\mathrm{Sp}(\mathbf{A}) = \{v_s - \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\ ,\ v_s + \sqrt{\frac{\sigma_0 n_s}{\rho_s \varepsilon_c^{n_s}} \varepsilon_s^{n_s - 1}}\},\quad \text{if } \varepsilon_s > \varepsilon_c$$

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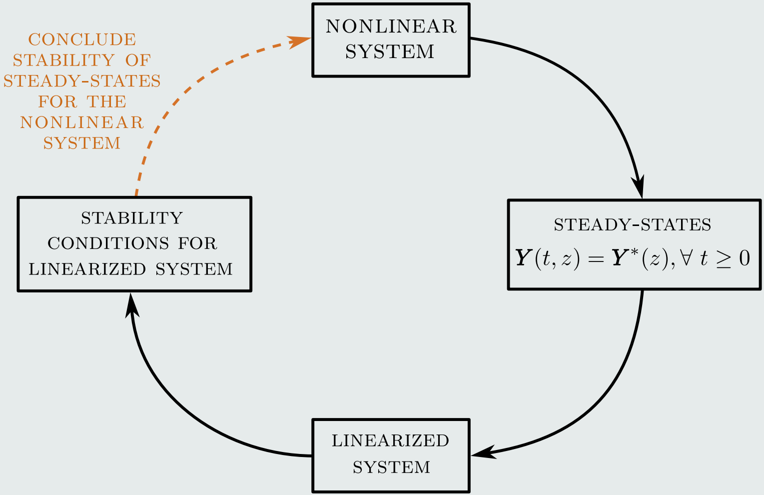
$\Rightarrow$ Weakly hyperbolic system

## System analysis

### How to stabilize ?



## System analysis: steady-states

Solve the following system of ODEs:

$$\begin{cases} \dfrac{\mathrm{d}f_s^*}{\mathrm{d}z} = 0, \\ \dfrac{\mathrm{d}\left(\frac{f_s^{*2}}{\varepsilon_s^*} + \frac{\sigma_e(\varepsilon_s^*)}{\rho_s}\right)}{\mathrm{d}z} = K\varepsilon_s^* - \gamma\dfrac{(\varepsilon_s^*)^\beta f_s^*}{1-\varepsilon_s^*}, \\ f_s^*(0) = 0, \quad f_s^*(z_b) = 0 \end{cases}$$

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For any given initial condition $\mathbf{Y}^0$ and fixed set of parameters ($\sigma_0$, $n_s$, $n_r$, $\varepsilon_c$, $\tau$, $\rho_s$), analytical expressions of system (1) steady-states are given by:

$$f_s^*(z) = 0, \qquad \forall z \in [0, z_b] \tag{I}$$

$$\varepsilon_s^*(z) = \begin{cases} 0, & 0 \le z < z_c \\ \left(\varepsilon_c^{n_s-1} + (n_s-1)\varepsilon_c^{n_s} g\left(\frac{\rho_s-\rho_l}{\sigma_0 n_s}\right)(z-z_c)\right)^{\frac{1}{n_s-1}}, & z_c \le z \le z_b \end{cases} \tag{II}$$

# Numerical simulation (Godunov-type Finite Volume scheme)

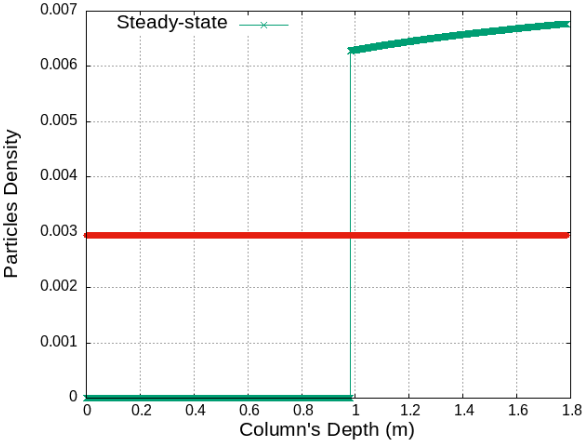


**Fig. Simulation of the particles density on 6400 cells using Godunov scheme with an LLF Riemann Solver for flux computation: comparison with steady-state**

## System analysis: Linearization

Consider deviations of the states with respect to steady-states:

$$E_s := \varepsilon_s(t,x) - \varepsilon_s^*(x), \qquad V_s := v_s(t,x) - v_s^*(x)$$

Linearized system is written as:

$$\partial_t \tilde{\mathbf{Y}} + \mathbf{A}^*(z) \cdot \partial_z \tilde{\mathbf{Y}} + \mathbf{B}^*(z) \cdot \tilde{\mathbf{Y}} = 0 \tag{2}$$

$$\tilde{\mathbf{Y}} = \begin{pmatrix} E_s \\ V_s \end{pmatrix}, \quad \mathbf{A}^*(z) = \begin{pmatrix} v_s^* & \varepsilon_s^* \\ \frac{\sigma_e'(\varepsilon_s^*)}{\rho_s \varepsilon_s^*} & v_s^* \end{pmatrix}, \quad \mathbf{B}^*(z) = \begin{pmatrix} \text{Remaining} \\ \text{source terms} \end{pmatrix}$$

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Rather transform $\mathbf{A}^*(z)$ into a *Jordan* block:

$$\mathbf{A}^*(z) \longleftrightarrow \begin{pmatrix} \mathbf{c}_1 & \mathbf{c}_2 \\ \mathbf{0} & \mathbf{c}_3 \end{pmatrix}$$

# Through a simplified clarifier

# Stability analysis of a simplified clarifier: a triangular hyperbolic system

## Target System

$$\begin{cases} \partial_t \phi(t,x) = \Lambda(x)\left(A(x)\partial_x \phi(t,x) + B(x)\phi(t,x)\right), \ x \in (0,1), t \geq 0 \\ \phi(t,1) = D\phi(t,0), \quad \textbf{(BCs)} \\ \phi(0,x) = \phi^0(x), \quad \textbf{(ICs)} \end{cases} \tag{3}$$

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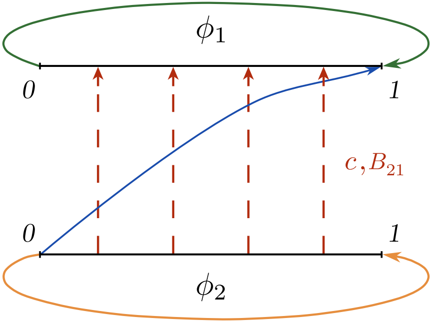


Schematic representation of the system (4)

# Stability analysis of a simplified clarifier: a triangular hyperbolic system

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$X = L^2((0,1);\mathbb{R}) \times H^1((0,1);\mathbb{R})$ equipped with the norm:

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### Exponential stability

The origin of system (2) is $\delta$-exponentially stable if there exists an overshoot $\kappa \geq 1$ and a decay rate $\delta > 0$ such that for any initial condition $\phi^0 \in X$:

$$\|\phi(t;\phi^0)\|_X^2 \leq \kappa e^{-\delta t}\|\phi^0\|_X^2, \ \forall t \geq 0 \tag{5}$$

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**To answer, stability analysis is performed. The study will be divided into two cases: Without and with source term B**

## Results: First case B = 0

Recall:

$$D = \begin{bmatrix} D_{11} & D_{12} \\ 0 & D_{22} \end{bmatrix}, \ \Lambda(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ 0 & \lambda_2(x) \end{bmatrix}$$

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### Assumption 1

The function $\lambda_2$ and the matrix $D$ satisfy the following inequality:

$$\max(D_{11}^2, D_{22}^2) < 1 - \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} \tag{6}$$

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### Theorem 1

Let Assumption 1 holds and let $\nu := \frac{\overline{\lambda_2'}}{\underline{\lambda_2} - \overline{\lambda_2'}}$, $\bar{\nu} := \frac{1-D_{22}^2}{D_{22}^2}$.

Then, $\forall \delta < \min\{\frac{\bar{\nu}\underline{\lambda_1}}{1+\bar{\nu}}, \frac{\bar{\nu}\underline{\lambda_2}}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\underline{\lambda_2}}{(1+\bar{\nu})(1+\nu)}\}$,

**The origin of system (4) is $\delta$-exponentially stable**

## Results: Second case B $\neq 0$

Recall:

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### Assumption 2

The function $\lambda_2$ and the matrices $B, D$ satisfy:

$$\max(D_{11}^2, D_{22}^2) < 1 - \max\left( 2\overline{B_{11}}, \; \overline{\frac{\lambda_2'}{\underline{\lambda_2}}} + 2\overline{B_{22}} \right) \tag{7}$$

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Let Assumption 2 holds and let

$$\nu_1 := 2\overline{B_{11}}(1 - 2\overline{B_{11}})^{-1}, \ \nu_2 := \frac{2\overline{\lambda_2'} + 2\underline{\lambda_2}\overline{B_{22}}}{\underline{\lambda_2} - 2\overline{\lambda_2'} - 2\underline{\lambda_2}\overline{B_{22}}}, \ \bar{\nu} := \frac{1 - D_{22}^2}{D_{22}^2}.$$

Then, $\forall \delta < \min\{\delta_1, \delta_2\}$ where $\delta_1 = \frac{\underline{\lambda_1}(\bar{\nu}-\nu_1)}{(1+\nu_1)(1+\bar{\nu})}$ and $\delta_2 = \frac{\underline{\lambda_2}(\bar{\nu}-\nu_2)}{(1+\nu_2)(1+\bar{\nu})}$, **the origin of system (4) is $\delta$-exponentially stable**.

## Sketch of the Proof:

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Consider the following functional:

$$\mathcal{V}(\phi) = \mathcal{V}_1(\phi) + \alpha_{20}\mathcal{V}_{20}(\phi) + \alpha_{21}\mathcal{V}_{21}(\phi)$$

With

$$\mathcal{V}_1(\phi) = \int_0^1 \tfrac{1+\mu x}{\lambda_1(x)} \phi_1^2(x) dx,$$
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Recall that $\mathcal{V}$ induces a norm on $X$ that is equivalent to the state norm $\|\cdot\|_X$ ∎

# Numerical results

# Example

Consider system (4) with:

$$\lambda_1(x) = \begin{cases} 10, & x < \frac{1}{2} \\ -4(x-\frac{1}{2})^2 + 10 & \frac{1}{2} \leq x \leq 1 \end{cases}$$

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$c(x) = x + 4$, $D = \begin{bmatrix} \frac{1}{2} & 1 \\ 0 & \frac{1}{2} \end{bmatrix}$, $B = \begin{bmatrix} \frac{x}{6} & 100x^3 \\ 0 & \frac{x}{6} \end{bmatrix}$, $\phi_1^0(x) = -\frac{2}{3}$,
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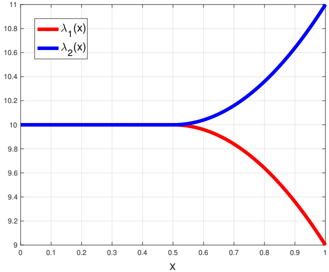

## Academic Example

Regarding Theorem 1, the assumption $\max(D_{11}^2, D_{22}^2) = \frac{1}{4} < 1 - \frac{\overline{\lambda_2'}}{\underline{\lambda_2}} = \frac{3}{5}$ is satisfied.

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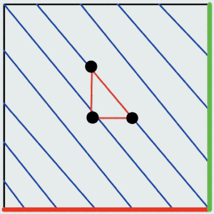
$$\begin{cases} \frac{\phi_{1,i}^{n+1} - \phi_{1,i}^{n}}{\Delta t} = \lambda_{1,i}\frac{\phi_{1,i+1}^{n} - \phi_{1,i}^{n}}{\Delta x} + c_i\lambda_{1,i}\frac{\phi_{2,i+1}^{n} - \phi_{2,i}^{n}}{\Delta x} \\ \frac{\phi_{2,i}^{n+1} - \phi_{2,i}^{n}}{\Delta t} = \lambda_{2,i}\frac{\phi_{2,i+1}^{n} - \phi_{2,i}^{n}}{\Delta x} \end{cases}$$

Characteristic lines through (x,t) plane

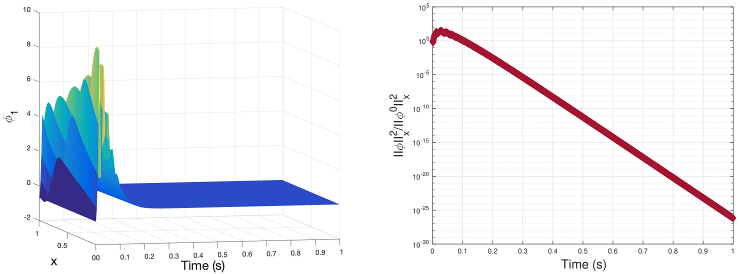


Setting:

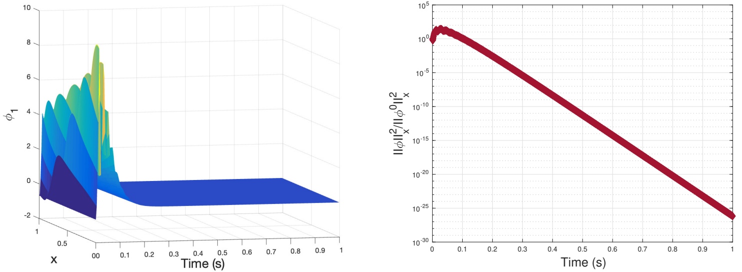
$$\gamma_{1,i} = \frac{\lambda_{1,i}\Delta t}{\Delta x}, \qquad \gamma_{2,i} = \frac{\lambda_{2,i}\Delta t}{\Delta x}$$

finally we get the numerical scheme:

$$\phi_{1,i}^{n+1} = (1 - \gamma_{1,i})\phi_{1,i}^{n} + \gamma_{1,i}\phi_{1,i+1}^{n} + c_i\gamma_{1,i}(\phi_{2,i+1}^{n} - \phi_{2,i}^{n})$$
$$\phi_{2,i}^{n+1} = (1 - \gamma_{2,i})\phi_{2,i}^{n} + \gamma_{2,i}\phi_{2,i+1}^{n}$$




**System (2) without source term: (left) state $\phi_1$ / (right) time evolution of $\frac{\|\phi\|_X^2}{\|\phi^0\|_X^2}$**

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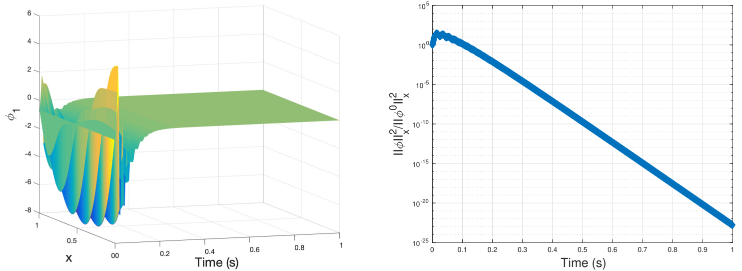


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Without source term, theoretically, decay rate is given by $\min\{\frac{\bar{\nu}\underline{\lambda}_1}{1+\bar{\nu}}, \frac{\bar{\nu}\underline{\lambda}_2}{1+\bar{\nu}}, \frac{(\bar{\nu}-\nu)\underline{\lambda}_2}{(1+\bar{\nu})(1+\nu)}\} = \frac{7}{2}$, whereas, numerically, decay rate is

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**The difference is due to the conservatism introduced by the proposed Lyapunov analysis**

# Conclusions and perspectives

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## Perspectives

### Ongoing work

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$$\begin{cases} \partial_t \phi(t,x) = \Lambda(x)\left(A(x)\partial_x \phi(t,x) + B(x)\phi(t,x)\right), \ x \in (0,1), t \geq 0 \\ \phi(t,1) = \begin{bmatrix} D_{11} & D_{12} \\ \mathbf{0} & D_{22} \end{bmatrix} \phi(t,0), \\ \phi(0,x) = \phi^0(x) \end{cases}$$

Tackle the case where the system **is not in cascade** form (relax coefficient $\mathbf{D}_{21}$)

$\Rightarrow$ Design a wider class of controllers

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$$\begin{cases} \partial_t \phi(t,x) = \Lambda(x)\left(A(x)\partial_x \phi(t,x) + B(x)\phi(t,x)\right), \; x \in (0,1), t \geq 0 \\ \phi(t,1) = \begin{bmatrix} D_{11} & D_{12} \\ \mathbf{0} & D_{22} \end{bmatrix} \phi(t,0), \\ \phi(0,x) = \phi^0(x) \end{cases}$$

Tackle the case where the system **is not in cascade** form (relax coefficient $\mathbf{D}_{21}$)

$\Rightarrow$ Design a wider class of controllers (of *Dynamical type* ?)

# Perspectives

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**e.g.**

$$\frac{d}{dt}\begin{bmatrix} \phi_1(t,1) \\ \phi_2(t,1) \end{bmatrix} = D \begin{bmatrix} \phi_1(t,0) \\ \phi_2(t,0) \end{bmatrix} + E \begin{bmatrix} \phi_1(t,1) \\ \phi_2(t,1) \end{bmatrix}$$

Where $D$ and $E$ are **full** matrices

**2)** Conclude about the stability of the nonlinear system

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**3)** Apply this theory to the Clarifier/Aeration tank configuration.

# Thank you for your attention
# Questions ?